

# SUFJAN FANA

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## EDUCATION

### University of California, Berkeley

June 2023 – December 2025

*Applied Mathematics, B.A. + Computer Science, B.A.*

- Relevant Courses: Machine Learning & Data Analytics, Deep Neural Networks, Computer Vision, User Interface & Design, Computer Architecture, Optimization Models in Engineering, Computer Security, Numerical Analysis

## EXPERIENCE

### Network Data Analyst

January 2025 – June 2025

*Berkeley Student Cooperative (BSC)*

- Documented and analyzed campus co-op networks across 10+ houses; inventoried 300+ devices (routers/switches/APs) and validated configs against documented standards
- Built Python scripts to clean/deduplicate logs before auditing; reduced house visit time by 1–2 hours on average
- Designed a standardized inventory schema and handoff template, improving cross-house record consistency

### Data Informatics Intern

August 2024 – December 2024

*Museum of Vertebrate Zoology, U.C. Berkeley*

- Wrote 20+ SQL queries and Python cleanup scripts to normalize locality fields; improved portal search recall and reduced invalid entries by ~30%
- Tested portal features end-to-end; filed 25+ actionable bugs with repro steps and authored Python/R/HTML scripts to automate weekly data pulls, cutting manual exports by ~4 hours/week
- Generated and published GIS layers/maps for 200+ specimens

### Research Assistant, Computational Linguistics

August 2024 – June 2025

*Alexandra Pfiffner, U.C. Berkeley*

- Automated phonetic segmentation with Python and R, reducing manual labeling time per recording
- Standardized ELAN/Praat annotation workflows, improving segmentation consistency and reuse
- Used scripting and plotting to analyze implosive acoustics and delivered publication-ready technical figures

### Information Technology Intern

January 2021 – August 2022

*Foothill-De Anza College*

- Resolved high-volume tech support tickets for 20,000+ students/faculty, improving resolution consistency
- Performed updates and software configuration across lab machines to improve reliability and performance
- Trained users on new tools and basic cyber awareness; wrote quick-start docs adopted across departments

## PROJECTS

### Diffusion Image Generation and Editing | Python, PyTorch, DeepFloyd IF

- Implemented diffusion noising/denoising with classifier-free guidance; reproduced SDEdit edits and RePaint inpainting on custom prompts
- Built/trained a conditional U-Net for MNIST denoising with class conditioning and timestep embeddings
- Benchmarked quality/compute tradeoffs across step counts and guidance scales; documented artifacts

### Steering Analysis for Vision-Language-Action Models | Python, PyTorch

- Reproduced activation-level steering via token-space clustering on OpenVLA; validated controllable behavior shifts with robot rollouts
- Compared token-projection steering vs. SAE probing on pio.5; analyzed feature geometry and entanglement
- Ran ablations over ~10M residual activations (normalization, layer choice); documented stability limits

### Automatic Panorama Stitching and Seam Blending | Python, NumPy, OpenCV

- Built a panorama pipeline: feature matching, robust homographies, image warping, and multi-image stitching for photo sequences
- Estimated homographies with DLT + RANSAC using Harris corners and ANMS; tuned thresholds to improve inliers
- Reduced visible seams with selection and multi-band blending; evaluated outputs via qualitative comparisons

## TECHNICAL SKILLS

**Languages:** Python, SQL, Java, Go, TypeScript, JavaScript, HTML/CSS

**Tools/Platforms:** Git, Linux, Docker, AWS, GitHub Actions, PostgreSQL, Redis

**Libraries/Frameworks:** FastAPI, React, Next.js, PyTorch, HF Transformers, OpenCV, NumPy, pytest